On the Germanic type *lūkan*

0. The Germanic class II strong verbs of the type Go. *lūkan* 'close, lock' (OIc. *lúka,* OE *lūcan* etc., Seebold 338) show an anomalous vocalism in Gmc. **-ū-,* beside the short diphthong **-iu-* otherwise characteristic of this class. The traditional explanation for this **-ū-,* although widely accepted, is unsatisfactory on several counts, and must be abandoned.

1. The *communis opinio* holds that the *-ū-* is analogical, having arisen under the influence of class I presents with Gmc. **-ī-,* from full-grade **-eį-* (thus e.g. Hirt, *Hdbch. des Urgerm.* 2. 169; Prokosch, *Comp. gramm.* 154; more recently, van Coetsem, *System* 69 and Orbis 12, 1963, 278, and further Ztschr. f. Mundartforsch. 29, 1962, 224–5, with bibliographical discussion.) There is less agreement as to what morphological category was remodeled into the *lūkan*-type, but the most popular version of the analogical approach assumes the source to have been zero-grade presents (so-called 'aorist presents') of the type Ved. *tudáti* (thus already Osthoff, PBB 8, 1881, 287 ff. and Streitberg, *Urgerm. Gr.* 292). On the basis, then, of class I forms like Go. *steigan* (with Gmc. **-ī-*) : *staig* : *stigum* 'steigen' etc., an original **lŭkan* (: *lauk* : *lukum*) was remodeled to *lūkan.*

Precisely the approach just sketched is promulgated most recently by Feuillet (BSL 76, 1981, 204), who flatly states (with no explanation): 'Comme toute explication de type phonétique est exclue, la règle du "modèle morphologique" est la seule qui permette de rendre compte de ce genre d'anomalie'. But the contention that 'a phonetic explanation of any kind is excluded' is simply false[1]).

2. Kuryłowicz opts for a somewhat different analogical approach, as follows (*Mél. Fourquet* 157 ff. = *Esquisses* 2. 382 ff.). According to Kuryłowicz, the source of the *lūkan* verbs is again a class of zero-grade presents. The remodeling of **lŭkan,* however, is based not on the class I presents, but on class III and IV verbs with root-shape *TReT-*. For such verbs, a secondary zero-grade in *TRuT-* is fairly well-attested, as in OHG *sprechan* 'speak', ptcple. *gisprochan, brechan* 'break', ptcple. *gibrochan* (cf. Go. *brukans,* OE *brocen*), etc. Thus the zero-grade of the class III/IV pattern (full-grade) *(T)ReT-* : (zero-grade) *(T)RuT-* coincides with the class II *tudáti*-presents of the shape *(T)RuT-* (beside class II full-grade presents *(T)ReuT-*); this situation leads to a differentiation of the class II zero-grades, by means of an 'insertion' of *u* comparable to the class III/IV pattern *(T)ReT-* : *(T)RuT-*. According to this schema, *lūkan* therefore represents a **luukan,* which is thus a morphological zero-grade replacing the inherited zero-grade type **lŭkan.*

3. Both of the theories just mentioned assign a pivotal role to a zero-grade type **lŭkan,* reshaped analogically to *lūkan.* But there are serious problems with this general scenario, beginning with the status of such zero-grade presents in Germanic. In the first place, one may wonder why there still exist a few apparent *tudáti* forms with *-ŭ-* which never underwent any such analogical remodeling (thus e.g. Go. *trudan,* OIc. *troþa* 'tread', with secondary full-grade in OE *tredan,* OHG *tretan,* according to the usual interpretation of these forms; see Kuryłowicz, *Mél. Fourquet* 160). Equally problematic is the existence of apparent *tudáti* forms with *-i-,* which might have been expected to be reshaped after class I full-grades even more readily than the **lŭkan* type (thus e.g. Go. **digan* 'knead', attested only in participial forms; note the pres. ptcple. *digandin,* Rom. 9: 20).

Also disquieting is the scarcity of direct evidence favoring an association between the *lūkan* verbs and a class of zero-grade presents. Streitberg's comparison of OE *būgan* etc. 'bend' with Ved. *bhujáti* 'folds, bends' and Gk. *φυγεῖν* 'escape, flee' (loc. cit.) is of limited value: to begin with, the IE stem-final **-ĝ-* presupposed by *bhujáti* does not accord with the **-gh-* of Gmc. **būga-*; the comparison with *φυγεῖν,* moreover, is still less certain, since the semantic relationship between *bhujáti* and *φυγεῖν* is thus far completely unclear (see the discussions by Mayrhofer s.v. *bhujáti* and Chantraine s.v. *φεύγω,* with further references). Otherwise, one finds in the literature only **sūka-* and **sūga-* 'suck', which are compared with the short-vowel zero-grade of OCS *sъsati* 'id.'. But the Auslautswechsel in **sūka-/*sūga-* is poorly understood (see Seebold 398 ff.), and the force of the comparison is weakened by e.g. Gmc. **liuga-* 'lügen' (Seebold 336), an ordinary class II full-grade present whose closest comparandum is again an OCS zero-grade present (*lъžati* 'id.'). The best (and perhaps the only) word-equation involving a *tudáti* form –apparently not noted in the literature– is Gmc. **lūka-* 'pluck' : Ved. *rujáti* 'breaks'. But this is a slender basis for the assumptions required by an analogical approach.

[1]) Ruijgh's variant of the standard approach (*Studies Palmer* 346) operates with an 'expressive *ū*', and will not be considered further.

4. Apart from the status of the zero-grade **lŭkan* type, there are several other difficulties with an analogical explanation, as Guxman et al. have observed (*Sravn. gramm.* 4. 304 ff.). Consider first the fact that many *ū*-verbs have counterparts throughout Gmc. (and sometimes within the same dialect) with regular full-grade vocalism *-iu-* (type OE *scūfan* 'shove' : OE *scēofan,* OHG *skioban,* Go. *-skiuban*; even for the *ū*-forms mentioned above with alleged zero-grade correspondents elsewhere, note e.g. Go. *biugan,* OHG *biogan* beside OE *būgan* and OHG *-liohan* (MHG *liechen*) beside the forms pointing to a Gmc. **lūka-* 'pluck'). Such a situation indicates that the source of *-ū-* is not likely to result from a unitary and relatively simple analogical process, which should result, rather, in constant *-ū-* if the original zero-grades were still functionally distinct from the ancestors of class II full-grade presents, or constant *-iu-* if the two types were no longer distinct.

Guxman et al. also note the existence of two further sets of data, neither of which is easily explained under an analogical theory: first, isolated (apparently zero-grade) *ū*-forms within Gmc., the best case being perhaps OIc. *þrútinn* 'swollen, oppressed', which has the appearance of being a participle to *þrjóta* 'come to an end, wear out' (itself with regular participle *þrotinn*); and second, extra-Gmc. comparanda with *-ū-*, of the type OIc. *súpa,* OE *sūpan* etc. 'saufen' (Seebold 399) : Ved. *sū́pa-* sauce, broth'.

On the basis of these considerations, Guxman et al. conclude that if *-ū-* was *inherited,* at least in some forms, then analogy with class I would account for the appearance of such etymological zero-grades in a category otherwise reserved for full-grades. Although it is necessary to assume, under any theory, that the *ū*-forms have spread in Gmc., there would be no need to explain the *genesis* of such forms secondarily (much less the isolated Gmc. forms like *þrútinn* or the forms with extra-Gmc. support for *-ū-*). But Guxman et al. proceed no further than the suggestion that the *-ū-* of some of the class II forms may be inherited.

Similar views were expressed (independently) by Schmitt-Brandt (*Vokalsystem* 67 n. 70):

> Daß die germ. Verben mit *ū* im Präsens wie got. *lūkan, lauk, lukum, lukans* 'schließen' nach Analogie zu *steigan* (*ī*), *staig, stigum, stigans* aus **lukan* entstanden ... ist unglaubhaft, da der Typus *liugan, laug, lugum, lugans* 'lügen' lautlich näher stand, also eher eine Neubildung **liukan* statt *lūkan* aus **lukan* zu erwarten gewesen wäre. Ist *lūkan* jedoch alt, so versteht sich leicht die Angleichung an den Typus von *steigan* in der Verteilung von *ū* : *u* wie *ī* : *i* im Verbalparadigma.

For the age of the type, note also von Kienle's more recent remark (*Histor. Laut- u. Form.*² 245) that the *-ū-* of *lūkan* etc. is 'eine altererbte Schwundstufe', although he comments no further.

5. If, then, the *-ū-* of such forms is inherited, it would result from set zero-grade formations, including analogical zero-grades in **(s)TRuH-(T-)* based on stems **(s)TREH-u-* or **(s)TERH-u-*. (Although Guxman et al. have nothing to say about the source and nature of this inherited *-ū-*, Schmitt-Brandt's comments – which are found in his discussion of the 'long-diphthong' roots – seem to imply just such an interpretation. The same possibility is admitted by Kuryłowicz, in his earlier treatment of certain aspects of the problem (*Apophonie* 317–19; note esp. 318, where Kuryłowicz attributes the same idea to Wissmann.))

It is conspicuous, in fact, that among six complexes of Gmc. verbs consisting of *d*-enlarged *u*-diphthongs based on stems *(s)TER(H)-* or *(s)TREH-*, five have parallel stems with *-ū-*:

a) As already mentioned, a potentially important form is the OIc. adj. *þrútinn* 'swollen, oppressed', evidently an isolated by-form of the regular pret. ptcple. *þrotinn* (to *þrjóta* 'become tired'), as Noreen suggests (*Altnord. Gr.* 328; similarly de Vries s.v., as a possibility). The same form is likely to appear in OE *aþruten* 'id.' (or perhaps 'inflamed, sore, wounded' or the like – cf. OE impers. *ā-þrēotan* 'it irks (me), it vexes (me)'); for the identification of OIc. *þrútinn* : OE *āþruten,* see Seebold 524, and further Holthausen 371 and Brunner-Sievers, *Altengl. Gr.* 297 [2]). If OIc. *þrútinn* : OE *āþruten* (*-þrūt-*) do indeed belong with Gmc. **þriuta-* (OIc. *þrjóta,* Go. *us-þriutan* 'verdrießen' etc., see Seebold 523–4), the pret. *ū*-vocalism is remarkable,

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[2]) Although the handbooks generally cite the form as *ā-þrūten,* or *ă̄-þrūten* (thus e.g. Bosworth-Toller, *Suppl.* 56; Brunner-Sievers and Holthausen, locc. cit.; Campbell, *OEG* 319–20; but Seebold *ā-þruten*), the two attestations listed by Bosworth-Toller are printed as *aþruten* (see Cockayne, *Leechdoms* 2. 44. 14 and 2. 218. 19), although the MSS for these texts are lightly accented. (See Cockayne's comments on the nature of the accentuation in the MSS, *Leechdoms* I. xciv ff.) Thus Brunner-Sievers voice a justifiable query as to whether or not *aþruten* is an error for *āþrunten* (pret. ptcple. to *þrintan* 'swell up'). A stem **þrūt-* is therefore not guaranteed for OE, but the equation with OIc. *þrútinn* is nevertheless attractive.

and leads one to consider the possibility of a class VII **þrūta-* : **þrūtana-* (thus Campbell, *OEG* 319–20 and Brunner-Sievers, loc. cit.).

b) A Gmc. **griuta-* is seen in traces of an OHG **-griozan* 'grind', found in two occurrences of pret. ptcples., Ahd. Gl. 2.207.67 *excollocta* : *fergrozzeniu* and 3.507.6 *panis cretinus* (i.e. *κρίθινος*) : *gigrozzan broth* (see Seebold 242); cf. further MHG *vergriezen* 'scatter, spill out', *durchgriezen* 'sprinkle (with sand)'[3]). The remaining Gmc. forms are nominal, in various words for 'sand, gravel, grit, grits'. Both full-grades and zero-grades are attested; for full-grades, e.g. *e*-grade **griuta-* (neut.) in OIc. *grjót,* OE *grēot,* OSax. *griot* 'gravel, sand' (OIc. also 'stone'), and *o*-grade **grauta-* (masc.) in OIc. *grautr* 'grits'[4]). The zero-grades, however, raise interesting problems. A short-vowel zero-grade **grut-* is attested in a neut. **gruta-* (OE *grot* 'coarse meal'), as well as in suffixed forms masc. **grut-ja-* (OHG (pl.) *gruzze-(melo)* 'grits') and fem. **grut-jō* (OE (pl.) *grytta* 'id.'). But there is also evidence for a consonant-stem **grūt-*, with long-vowel zero-grade, which is likely to be the most archaic of the various nominal forms (e.g. OE (f.) *grūt* 'coarse meal', MHG (m.) *grūz* 'sand, grain', Norw. (n.) *grút* 'grounds'.) There is also the possibility that the form does not continue an old root-noun, but represents a secondary transfer to the consonant declension (cf. Meid, *Germ. Sprachwiss.* 3.57). But in either case, the form would provide important evidence for a long-vowel stem **grūt-*, which may be old. In contrast, the short-vowel zero-grades can all reflect deverbative formations to the strong verb **griuta-*, and therefore do not necessarily support the existence of an old **grut-* in Gmc. Similarly, the ptcple. **(-)grutana-* itself (cf. the OHG glosses above) could have been formed automatically to **griuta-*, and need not show an old short-vowel zero-grade.

c) A more straightforward *lūkan*-type form is found in Gmc. **slūta-* (OFr. *slūta*), beside the regular full-grade **sliuta-* (OHG *sliozan* 'schließen' etc., Seebold 436–7). Although the outcome of initial **skl-* in Gmc. is uncertain (see § 6(d) below for the presumed basis **(s)kleh$_2$-u-*), the development **skl-* > **sl-* can be paralleled (cf. **slīta-*, § 8(c) below), and may well reflect the regular treatment

[3]) Pokorny's gloss 'in kleine Teile zerreiben' for *ver-, durchgriezen* has no support from the MHG lexica.

[4]) The *o*-grade adj. **grauta-* 'great, thick' (OE *grēat,* OHG *grōz* etc.) is usually placed here, although the semantic history of this word is unclear.

d) Similarly, beside Gmc. **spriuta-* 'sprießen' (OE *-sprēotan* etc., Seebold 459–60) a long-vowel **sprūta-* is also attested (note Seebold's entry *SPRŪT-A-*, in favor of *SPREUT-A-*). Du. *sproeten* 'spritzen' (cf. MLG *sprōte* 'spot, freckle') is generally interpreted as an *o*-grade (thus **sproh$_1$-d-*; see further below for the seṭ stem), but this is not certain. Note that the Low German reflexes of Gmc. **-ō-* also appear (dialectally) for **-ū-*: e.g. Du. *roest* : OE *rūst* 'rust', where a *u*-diphthong is assured (however the problematic–but in this case clearly secondary–OE *ū*-vocalism is to be explained). Since **sprūta-* is otherwise attested in Gmc. (esp. Low German), the vocalism of *sproeten* etc. can equally well reflect **-ū-*, as favored by van der Meer, *Hist. Gr. d. niederl. Spr.* 1.53–4.

For the fifth such item (**hriuta-/*hrūt-* 'fall'), the *ū*-stem appears in a weak verb; for configurations of this sort, see § 9 below.

6. The evidence for seṭ bases for these forms is as follows:

a) Gmc. **þriuta-/*þrūt(a)-* belongs with **terh$_2$-* 'pierce, bore, rub through' (Pokorny, *IEW* 1071ff.[5])), and more specifically its *u*-extended stem **terh$_2$-u-* (Hitt. *tarḫuzzi* 'overcomes', *τερύσκετο· ἐτείρετο* Hsch. etc.), which in fact has a zero-grade **truh$_2$-* (*τρύω* 'rub through', OCS *tryjǫ, tryti* 'rub' etc.). There is, moreover, a relatively extensive set of *d*-enlarged forms outside Gmc. (see *IEW* 1095–6), most of which are either ambiguous with regard to the presence or absence of a laryngeal, or clearly aniṭ (hence Pokorny's *tr-eu-d-, IEW* 1095). Thus Lat. *trūdō, -ere* 'thrust, shove, press' is ambiguous (as is Alb. *treth* 'cut, castrate', despite Meyer's **treu̯dō,* 435), but Lat. *trudis* (f.) 'pole, pike' is descriptively aniṭ[6]). The Celt. and Sl. material also appears to be aniṭ, e.g. W. *cythrudd* 'affliction, distress'[7]), OCS *trudъ* 'work, trouble' (SCr. *trûd*), *tružd*ǫ *sę, truditi sę* 'work hard, trouble oneself' etc. But Ba. shows an apparent **trūd-s-* in e.g. Li. *triūsas* (older *trūsas*) 'toil, labor', *tr(i)ūsti* 'to toil' etc.[8]).

[5]) This root may be the same as **terh$_2$-* 'cross over' (*IEW* 1074ff.); see Joseph, *IE laryngeals in Celt.* 188 for discussion.

[6]) The form does not appear until Virgil, and has been taken as a deverbative Neubildung to *trūdō* since Stolz (*Histor. Gr.* 1.454, followed by Solmsen, *Beitr.* 165 and Walde-Hofmann s.v. *trudis*; analyzed as deverbative most recently by Jirgensons, *Formal anal.* 291). But Lat. f. *i*-stems are not regularly created in this fashion, and the form may preserve an aniṭ zero-grade **trud-*.

[7]) According to *GPC* s.v., from Pr.-Celt. **kon-treu̯-d-* (*e*-grade is also possible); but there is no evidence that Celt. distinguishes *CEU̯*-C- and *CEU̯H*-C-, see Joseph, *IE laryngeals in Celt.* 372ff.

[8]) See Fraenkel s.v. *triūsas*; Latv. *trûdêt* 'cause to rot' is in all probability a productively-formed causative in *-dêt,* and does not support the **trūd-* of Li. *tr(i)ūsas.*

Although Ba., to be sure, creates zero-grades in *-ū-* fairly readily, Ba. **trū-d-* is not associated with any full-grade forms (Li. **tráud-*, or the like), and is difficult to dissociate from BSl. **trū-* (OCS *tryti*; Li. *trūniù, trūnėti* 'rot, decay' etc.).

b) Beside Gmc. **griuta-/*grūt-*, one likewise finds BSl. material with dental enlargement consistent with a set interpretation. Thus Li. *grūdžiu/grūdu, grūsti* (Latv. *grûžu/grûdu, grûst*) 'stamp, crush, push' could reflect an original **ghruh$_x$-d-*, comparable to Gmc. **grūt-* (cf. OPr. *grūdas* 'corn'). As just noted, however, such Ba. zero-grades can be secondary: witness Li. *plūstu, plūsti* (pret. *plūdau*) 'stream, flood' to IE **pleu̯-*[9]). But the vocalism and intonation of the Sl. forms point to set full-grades: CS *gruda* 'heap, pile' (Russ. *grúda*, SCr. *grȕda* 'id.'), SCr. *grȕmēn* beside Russ. *grum* 'clod, glebe' (see Berneker 1.357)[10]).

External evidence in favor of a set basis **ghreu̯h$_x$-* (or conceivably a *u*-basis **ghreh$_x$-u-*) is largely circumstantial, although suggestive. Pokorny, despite his **ghreu-d-*, nevertheless registers the *d*-forms just discussed under his 'long-diphthong' entry 2. *ghrēu-* : *ghrəu-* : *ghrū-* 'scharf darüber reiben, zerreiben' (*IEW* 460ff.), said to be an enlarged form of 2. *gher-* 'hart worüber streichen, reiben' (*IEW* 439–40). While e.g. Gk. **χραύω* (*ἐνέχραυε* Herod. 6.75, *χραύσῃ* Il. 5.138) 'scrape, graze, wound slightly' could reflect a full-grade of a *u*-basis, this interpretation depends on the connection with *χρώς* 'skin' (see Frisk s.v. *χρόα*, and Hamp, MSS 38, 1979, 84). But this connection is uncertain, and the morphology of *χρώς* is not entirely clear (see Szemerényi, SMEA 2, 1967, 21ff.; for **χραύω* : *χρώς*, note also Peters' analysis of Att. *τραῦμα* : Ion., Dor. *τρῶμα* 'wound', *Lautgesch. u. Etymol.* 328ff.). The remaining forms are either unusable, or ambiguous in various ways. Thus the lengthened-grade of

[9]) I do not believe that forms such as the Sl. infin. **plāti* (Russ. *plyt'*, SCr. *plȉti*) or OIr. *lúth* 'motion, vigor' (cf. *lúithir, lúaidid* 'moves') provide evidence for a set variant **pleu̯h$_x$-*, nor that the forms with *ō*-vocalism (e.g. Go. *flodus*, OE *flōd* etc. 'flood'; OIc. *flóa*, OE *flōwan* '(over)flow'; Gk. *πλωτήρ* 'sailor', *πλώω* 'sail') support a *u*-basis **pleh$_x$-u-*; see my *IE *-d-* 26, 42–3 for discussion.

[10]) It is not certain whether Li. *grúodas* 'hard frost, hard-frozen mud; malanders' belongs here (with Fraenkel s.v.), or, as traditionally, with the word for 'hail' seen in Lat. *grandō*, OCS *gradъ* (SCr. *grâd*) etc. (see e.g. *IEW* 406). In addition, certain Sl. forms (which do not agree with the acute intonation of either SCr. *grȕda, grȕmēn* or Li. *grúodas*) are claimed to show an 'Übertragung aufs seelische Gebiet' (Berneker 1.358, echoed by Pokorny, *IEW* 460–1), e.g. RCS *sъ-grustiti śa* 'grieve', OCS *grъdъ* 'proud' (SCr. *grd*) etc. But the uncertain semantics involved render such forms untrustworthy.

Pokorny's **ghrēuno-* (as in e.g. OIc. *grjón* 'grits', MHG *grien* 'gravel, sandy shore') is purely conjectural; and the Gmc. forms with meanings 'shudder, fear, frighten' and the like (see *IEW* 460–1) are semantically too distant to be used as evidence. If Jasanoff's analysis of the Gmc. Verschärfung is correct (see MSS 37, 1978, 77ff.), Norw./Ic. *grugg* 'grounds' (representing Gmc. **gruwwa-*) could be taken as evidence for a set stem; but the zero-grade would be analogical in any case (for Gmc. **-uww-*, **-ijj-*, see Jasanoff, op. cit. 86–7), and the corresponding full-grade form with Verschärfung (which is not attested) would not distinguish between an original set **ghreu̯h$_x$-*, and a secondary full-grade based on a zero-grade **ghruh$_x$-*, to an original *u*-basis **ghreh$_x$-u-*[11]).

c) For **sliuta-/*slūta-*, a *u*-basis **(s)kleh$_2$-u-* is in fact attested with some certainty (Dor. *κλᾱ(Ϝ)ίς* 'bolt, key', OCS *ključь* 'key' (SCr. *kljȕka*); see *IEW* 604–5 and Schmitt-Brandt, *Vokalsystem* 71), cf. **þriuta-/*þrūt(a)-* to **terh$_2$-u-*. Here again, another *d*-enlarged stem is quotable, namely Lat. *claudō, -ere* 'close'. This form is traditionally interpreted as **klāu̯i-d-e/o-* (e.g. Pokorny, Walde-Hofmann s.v., cf. Leumann, *Lat. Gr.* 1[2].97 for **-āu̯i-*), which seems to assume denominative origin, after *clāuis* 'key, locket'[12]). But *claudō* can also reflect **kléh$_2$-u-d-e/o-* directly, or else an analogical full-grade **kláu̯h$_2$-d-e/o-*[13]). Lat. *clūdō* (= *claudō*) is a patently late decompositional formation (cf. *in-clūdō, ex-clūdō* etc.)[14]).

d) Superficially comparable to Gmc. **spriuta-/*sprūta-* are cognate Ba. dental stems, once again, as in Li. *spráudžiu, spráusti* 'thrust, foist' beside *sprūdinti* 'cause to slide out', Latv. *sprūdît* 'release,

[11]) Also inconclusive are certain forms claimed to reflect a parallel **ghreh$_x$-i-* (IEW 457: *ghrēi-* : *ghrəi-* : *ghrī-* 'darüberstreichen, hart darüberstreifen, bestreichen (auch z.T. beschmieren, Schmutzstreifen')): e.g. Gk. *χρῑ́ω* 'rub, anoint, smear', *χρῖμα* 'salve' etc. provide evidence only for a zero-grade, and the Gmc. forms (e.g. OIc. *gríma* 'cowl, face-mask') are phonologically ambiguous in familiar ways, and in many cases semantically opaque. A further problem is Li. *griejù, griėti* 'skim off cream from milk', whose intonation is deviant under the assumption that the phonologically regular outcome of *U*-basis reflexes in Ba., in this position, is acute intonation.

[12]) Note that in this case, **klāu̯i-dh-* is equally possible.

[13]) For stems of the type **klau̯h$_2$-d-*, see my discussion of Lat. **caudō/cūdō* and related forms, *IE *-d-* 74–6.

[14]) Other dental forms sometimes adduced are semantically difficult, and cannot be connected to **sliuta-/claudō* with any certainty, namely RCS *kljuditi* '*λέγειν, ὁμιλεῖν*' (see Berneker 1.527 and Sreznevskij, *Materialy* 1.1229), and Lat. *claudus* 'lame' (see Walde-Hofmann s.v.).

push'[15]). But such forms are not decisive for set analyses **spreu̯h₁-d-* / **spreh₁-u-d-* and **spruh₁-d-*; in addition to forms like Li. *plūstu* (stem *plūd-*) to IE **pleu̯-*, cited above, note also Li. *pláudžiu* 'wash'. Although the basis **sper(h₁)-* 'strew, sow; spray' (*IEW* 993–5) is predominantly anit, Joseph points out (*IE laryngeals in Celt.* 181) that Gmc. forms like OHG *sprāt* 'Spritzen' require a **spreh₁-*[16]), which will account as well for certain Celt. forms (e.g. OIr. *sráb* 'stream; multitude', either from full-grade **sproh₁-bho-* or zero-grade **spr̥h₁-bho-*).

7. For most of the above Gmc. dental forms, then, there is good evidence at least for the set character of the diphthongal stem: for **þriuta-* / **þrūt(a)-* and **sliuta-* / **slūta-*, *u*-bases **terh₂-u-* and **(s)kleh₂-u-* are well-attested; for **griuta-* / **grūt-*, note the set stem presupposed by Norw./Ic. *grugg* 'grounds', and perhaps also by Gk. **χραύω* 'scrape'. Here one may also add the remaining Gmc. *d*-enlarged verb based on a stem in *u*-diphthong, namely **briuta-* 'break' (OIc. *brjóta*, OE *brēotan* 'break, destory', see Seebold 141–2, also 128 s.v. BRAUT-*A-), cf. RV *bhrūṇá-* 'embryo' and the Ba. acute in Latv. *braũna* 'scab, scurf' and Li. *br(i)áutis* 'push on, advance'[17]). It is thus reasonable to suppose that the Ba. dental comparanda do reflect, in general, set stems, although some forms may of course represent late or productive formations with lengthened grade, or with secondary long-vowel zero-grades. Thus the length of Li. *grū́džiu* 'crush' (: **griuta-* / **grūt-*, note also SCr. *grȕda* 'heap') and Li. *triūsas* 'toil' (: **þriuta-* / **þrūt(a)-*) may be old, but Li. *spráudžiu* 'thrust', *sprū́dinti* 'cause to slide out' may be secondary, especially since there is no independent evidence for a set stem with *u*-diphthong to this root (whether **sperh₁-u-*, **spreh₁-u-*, **spreu̯h₁-*, or zero-grade **spruh₁-*).

For the *lūkan* type in general, moreover, it is noteworthy that such Ba. comparanda are relatively frequent, thus **liuka-* / **lūka-* 'pluck': Li. *láužti* 'break (trans.)', *lū́žti* 'break, split (intr.)'; **riuka-* / **rūka-* 'rauchen' : Li. *ráugti* 'leaven; pickle', Latv. *rûgt* 'pickle, smoke' (for the semantics, see Seebold 380); **smiuga-* / **smūga-* 'schmiegen' : Li. *smáugti* 'strangle, choke', or perhaps *smùkti* 'slip, fall down', *smáukti* 'draw, pull (off)' (see Seebold 440). While such forms can occasionally be found beside ordinary class II forms in *-iu-* (with no alternant *-ū-*), one should also note that many of the regular class II forms have transparent etymologies with anit diphthongal stems, of the type **fliuga-* / **fliuta-* 'fliegen/fließen' : **pleu̯-*, **kiusa-* 'choose' : **ǵeu̯s-*, **riuda-* 'röten' : **h₁reu̯dh-*, etc. Etymologies of this sort are conspicuously absent from the *lūkan* verbs (with very few possible exceptions), *including* those with a regular full-grade variant.

Thus the *lūkan* type – or at least the subset of *lūkan* verbs with *d*-enlargement, which seems to form the most archaic core – may well have its source in zero-grades *TRuH-* to *u*-bases *TERH-u-* or *TREH-u-*, the oldest case perhaps being **terh₂-u-*, where the *u*-basis is clearly inherited, and where the Gmc. evidence for **þrūt(a)-* is isolated and possibly archaic. In accordance with Schmitt-Brandt's observations (see § 4 above), it is easy to understand how an inherited non-alternating (pres.) **þrūt-* : (pret.) **þrūt-* could have been reshaped to **þrūt-* : **þrŭt-*, with the old pret. stem **þrūt-* ousted from the paradigm, and the ptcple. lexicalized as an adj. The unstable type **þrūt-* : **þrŭt-* was then liable to acquire a full-grade present **þriut-*, as is often found, we have seen, beside the original long-vowel present. The process has clearly spread far beyond its original bounds, such that forms like **spriuta-* / **sprūta-* by no means imply an inherited *u*-basis **spreh₁-u-* or **sperh₁-u-*. (See §§ 9 and 10 below for further remarks on inner-Gmc. developments.)

It should be noted in passing that this analysis of the *lūkan* type serves to locate, within Gmc., the otherwise well-known type exemplified by Gk. *τρῑ́βω* 'rub' etc., involving precisely (thematic) enlarged formations based on (analogical) zero-grades *TRUH-* to *U*-bases *TERH-U-* or *TREH-U-*. It should also be clear that traditional (Perssonian, Pokornian et al.) analyses of the type *(s)TR-eu̯-d-* (see below for *(s)TR-ei̯-d-*) have no historical validity: the **-d-* of the Gmc./BSl. forms discussed above does not in fact occupy the second enlargement position, and there is no need to account for an otherwise puzzling profusion of forms with first enlargement **-eu̯-* (or **-ei̯-*). But the precise history of the development of 'long-diphthong' stems in the Gmc. verb – which are now seen to have extended beyond class VII – remains to be written.

8. One could also wonder whether at least some Gmc. class I presents reflect the same formation. This query was answered in the

[15]) For a discussion of the semantics of **spriuta-* and *spráudžiu*, see Seebold 459. A trace of **(s)prūd-* has been seen in Sl., cf. Russ. *pryt'* (*vo vsju pryt'* 'as fast as one can (run)'), *prytkij* 'quick, nimble', Pol. *pryciać* 'hurry' etc., Vasmer 2.453. (For the 'secondary' Sl. *s*-mobile, cf. Stang's comments on RCS *cěditi* 'strain, sift' (allegedly to **sḱeh₂-i-* 'cut'), *Lex. Sonderübereinst.* 86.)

[16]) Cf. also MHG *sprœjen, sprœwen* 'fly about (of sparks etc.)', from Gmc. **sprēwjan.*

[17]) For more detailed discussion of these and other related forms, see my remarks in *IE* **-d-* 54–6.

negative by Kuryłowicz (*Mél. Fourquet* 160, 160 n. 1), but such a situation is inherently unlikely. Consider the following class I presents of the shape *(s)TRīta-* with etymological *d*-enlargement:

a) A Gmc. **drīta-* appears in OIc. *dríta* 'defecate', and traces pointing to OE **drītan,* OHG **trīzan* 'id.' (for further forms and discussion, see Seebold 164). A dental stem is also attested in Slavic, but with forms pointing to both **-d-st-* (or **-d-t-*) (e.g. Russ. dial. *dristát'* 'have diarrhea', cf. Vasmer 1.370), and **-d-sk-* (e.g. SCr. dial. *dríškati* 'defecate noisily'), on which see Berneker 1.224. For the root **dher-* 'defecate; filth', see *IEW* 256, and perhaps also **dher(h₂)-* 'stir up (grounds); grounds, dregs', *IEW* 251–2[18]).

b) The Go. hapax *disskreitands* (Mk. 14: 63) points to a Gmc. **skrīta-* 'tear up (trans.)', beside the intransitive *disskritnan* 'become torn' (also poorly attested: 3 sg. pret. *disskritnoda* Mt. 27: 51 and Mk. 15: 38, 3 pl. pret. *disskritnodedun* Mt. 27: 51)[19]). For *(s)ker(hₓ)-* 'cut', see *IEW* 938 ff., and the useful survey by Joseph, *IE laryngeals in Celt.* 156 ff.

c) Gmc. **slīta-* (OIc. *slíta* 'slit, break', OE *slītan* 'tear up', OHG *slīzan* 'split, tear' etc., Seebold 430–1) belongs with IE *(s)kel(hₓ)-* 'split, cut' (*IEW* 923 ff.), assuming the treatment **skl-* > Gmc. **sl-* for the initial (cf. **slūta-,* § 5(c) above)[20]).

[18]) Other forms traditionally connected with Gmc. **drīta-* and Sl. **drist-/*drisk-* are problematic. The old comparison with Gk. *θρίδαξ* 'wild lettuce', still mentioned by Seebold, is completely unverifiable (note the variants *θρύδαξ, θρόδαξ* and others). Li. *tríedžiu, tríesti* 'have (a bad case of) diarrhea' (Miklosich 50, retained by Seebold) has a deviant initial (pointing simply to **tr-,* not **dhr-*), and can in fact be assigned to a different root altogether (see Fraenkel s.v. *tríesti* 2). Seebold (164–5) suggests that the following representative forms are all primordially related to one another: OIc. *dríta* etc.; OE *tord,* OHG *zort* 'excrement'; Li. *tríedžiu*; Lat. *stercus* 'excrement', with the 'Wurzelanlautsvariante' **dher-, *der-, *ter-* and **ster-* concealing a primitive identity. While material of this sort is an easy target for distortions and deformations, Li. *tríedžiu* would add nothing significant beyond its intonation, for which cf. the remarks in § 7 above on Li. *brántis,* and on the frequency of Ba. comparanda with acute intonation beside *lūkan* forms.

[19]) An *s*-less variant Gmc. **krīta-* may appear in OSax. **hrītan* 'scratch, write' etc. (Seebold 272); but the analysis of these forms, which may well belong with **wrīta-* 'id.' (see § 8(h) below), is highly uncertain. OIr. *scrissid* 'scraper' may point to an anit zero-grade stem **skri-d-* (thus Pokorny, *IEW* 946); but a zero-grade **skr̥-d(h)-* is also possible, as preferred by Vendryes S-39 (similarly Joseph, *IE laryngeals in Celt.* 160).

[20]) Ba. cognates are sometimes seen in Li. *skleidžiù, skleīsti* 'spread out, open up (trans.), burst' and similar forms (see Fraenkel s.v. for additional Ba. material). But Seebold (431) has observed that the meaning of *skleīsti* is somewhat

d) Gmc. **splīta-* 'split (trans.)' appears to belong with IE *(s)pel-* 'id.' (*IEW* 985 ff.; for the problem of the voiceless aspirate initial –hence Pokorny's "(s)p(h)el-"–see Hiersche, *Tenues aspir.* 72–3, 149 ff., 192 ff.); for the most part, however, the forms are attested relatively late. Aside from OFr. *splīta* 'split, splice', Seebold (454) records only MLG *spliten,* Du. *splijten* 'split', MHG *splīzen* 'spleissen'. Otherwise, one finds only MHG *splitter* 'splinter' and similar forms (e.g. an apparent denominative in MLG *splitteren* 'zersplittern'), and a series of (mostly late) forms with nasal, both with and without *s-*, and with variation in the stem-final dental, as reflected in Eng. forms like *splint, splinter, flint, flinders* etc. (see *IEW* 1000). Although the source of the nasal is unclear, one need not begin from an archaic nasal present.

e) Beside **spriuta-/*sprūta-* (to **sper(h₁)-* 'strew, sow; spray', *IEW* 993–5), there exist a few forms with *i*-diphthong, showing both **-d-* and **-t-*, as in OHG *sprīzan* 'splinter; sprinkle' beside MHG *sprīden* 'spread out, scatter, split up (intr.)'.

f) The isolated OE *þwītan* 'shave off (a splinter), chop off' is traditionally compared with Li. *tvóju, tvóti* 'thrash, cudgel, throw down' (see Pokorny's tentative lemma **tu̯ĕi-* 'scharf hauen, schlagen', *IEW* 1099). Note that if the *je/o*-stem of *tvóju* in fact continues a 'long-diphthong' *i*-verb (as in the type OCS *spějǫ* 'thrive' to **speh₁-i-*), the Ba. vocalism points to **tu̯eh₂-i-*, in which case *þwītan* could reflect a zero-grade **tu̯ih₂-d-* or a secondary full-grade **tu̯eih₂-d-*, but not a full-grade **tu̯eh₂-i-d-* (> Gmc. **þwaita-*). See further § 10 below for further remarks on secondary full-grade interpretations.

g) An isolated Gmc. **wlīta-*, together with an *o*-grade **wlaitō-*, are apparently based on **u̯el-* 'see' (*IEW* 1136–7); for OE *wlītan* 'see, show', Go. *wlaiton* 'look for' etc., see Seebold 563–4.

h) Pokorny assigns Gmc. **wrīta-* 'scratch, write' (OE *wrītan* etc., Seebold 566–7) to his 7. *u̯er-* 'aufreissen, ritzen' (a 'Grundlage für Erweiterungen', see *IEW* 1163–4). The same (or a similar) stem may appear in Arm. *ergic-anem,* aor. *ergici* (but normally causat., *ergicuc'anem* 'tear up, break up'), which could reflect a **u̯r-(e)i-d-s-*, according to Frisk (GHÅ 50, 1944, 28 ff. = *Kl. Schr.* 276 ff., cf.

deviant (from Gmc. **slīta-*); in fact, it can be said that the general meaning of Li. *skleīsti* involves merely extension in space (see e.g. Niedermann-Senn s.v.), and that usages indicating violent action ('burst' etc.)–and therein lies the semantic basis of the comparison with Gmc. **slīta-*–are incidental, secondary extensions. The etymological connection can therefore be doubted.

Pokorny's mechanical **u̯r-e̯i̯-d-* for the ambiguous **wrīta-*), although stem-final **-d-i̯(e/o-)* is preferable (see Godel, *Introduction* 82)[21]).

Many of the above Gmc. forms, then, are parallel to **griuta-/*grūt-*, **þriuta-/*þrūt(a)-* in certain important respects; or more precisely, they are comparable to the type **spriuta-/*sprūta-*, in that they are based on *(s)TER(H)-* roots without strong evidence for *i*-bases *(s)TERH-i-* or *(s)TREH-i-*. In one case, there is in fact some evidence for a zero-grade *(s)TRiH-*: thus for **wrīta-*, Gk. ῥῑ́νη 'file, rasp; shark', ῥῑνός 'hide' point to a seṭ zero-grade **u̯rih$_x$-*; but the relationship of such a form to the alleged basic **u̯er-* (or even **u̯erh$_x$-*) is completely unclear, nor is there any way of knowing whether the corresponding full-grade was a **u̯reu̯ih$_x$-*, or even an *i*-basis **u̯reh$_x$-i-* (whence analogical zero-grade **u̯rih$_x$-*). In two other cases, an *i*-basis interpretation is possible, but not certain: thus for **skrīta-*, Lat. *scrībō* 'scratch, write' (with ambiguous stem vowel) and Gk. σκαρῑφάομαι 'scratch, scrape' may conceal a **skerh$_x$-i-* or **skreh$_x$-i-* (although the Gk. form shows an unexplained vocalization of the presumed analogical zero-grade **skrih$_x$-bh-*); and Gmc. **þwīta-* may continue a zero-grade or secondary full-grade based on a **tu̯eh$_2$-i-*, cf. Li. *tvóju*. In general, however, evidence for zero-grades in *(s)TRiH-* (much less *i*-bases *(s)TERH-i-* or *(s)TREH-i-*) is simply lacking (contrast well-attested **terh$_2$-u-* and **truh$_2$-* beside **þrūt(a)-*). One may therefore assume that the type **drīta-*, **skrīta-* etc. is in general later than the type **sprūta-*, **slūta-* etc. The limited semantic range of the rhyming set **skrīta-* 'cut', **slīta-* 'slit', **splīta-* 'split', **þwīta-* 'shave off', **wrīta-* 'scratch, write' is also noteworthy, and is likely to result from late and secondary developments. The form **sprīta-* itself appears to be a secondary variant of **sprūta-*, and has no extra-Gmc. support; cf. also **wlīta-* 'see', without the slightest external support for a diphthong of any kind.

In fact, it is worth pointing out that traditional analyses of these stems in terms of a doubly-enlarged *(s)TR-e̯i̯-d-* derive no support from alleged 'intermediate' stems *(s)TR-e̯i̯-*, which turn out to be largely illusory, as follows:

[21]) For Arm. *ergic-*, even **u̯reh$_1$-d-i̯/s-* is a theoretical possibility, cf. OPr. *redo* 'furrow', older Dan. *vraade, vrode* 'dig, root' ("**u̯rēd-*"), as well as OE *wrōtan*, OIc. *róta* etc. 'dig, root' ("**u̯rōd-*"); note further an aniṭ **u̯erd-* in OCS *vrěda* (Russ. *véred*) 'wound', with descriptive Schwebeablaut in RV *ávradanta* 'they became soft' (see Anttila, *Schwebeablaut* 155–6). The relationship of these stems to one another (and to **u̯rih$_x$-*, see below) is quite unclear.

i) For an 'intermediate' *(s)kr-e̯i̯-* (to *(s)ker(h$_x$)-* 'cut', cf. Gmc. **skrīta-*); the only real evidence comes from the **kre̯i̯-* of Gk. κρῑ́νω, Lat. *cernō* 'separate, discern', OIr. *criathar* 'sieve' etc.; but it is uncertain whether this **kre̯i̯-* 'separate, sift' is to be analyzed diachronically as **(s)kr-e̯i̯-* (Joseph, op. cit. 167, tentatively sets up a **kreh$_1$-i-* for forms like Lat. *crēuī* 'separated', OIr. *crích* 'border', as well as Gk. κρῑ́νω, Lat. *cernō* etc. Although the aniṭ forms of the latter group (note e.g. ptcple. κριτός) are not easy to motivate, the analysis has certain advantages for the former group; but even if it is correct, **kreh$_1$-i-* 'separate, sift' need not be related with **(s)ker(h$_x$)-* 'cut')[22]).

ii) Beside Gmc. **slīta-*, Pokorny sees a **skl-e̯i̯-tro-* in OIc. *slíðr* (n.pl.) 'lath (for making scabbards); sheath, scabbard'. But the vocalism of Gmc. **slīþra-* is as ambiguous as that of **slīta-*. The alleged Ba. *s*-less forms (Li. *klajóti* 'wander about', Latv. *klajât* 'id.', see Fraenkel s.v. *klýsti*) seem to bear little semantic relationship with **slīta-*, and are regarded with suspicion even by Pokorny. Indeed, the vocalism of their dental pendants (Li. *klýstu, klýsti* (pret. *klýdau*) 'lose one's way', cf. Latv. *klîstu* 'wander') points to seṭ **(s)klih$_x$-d(h)-*, unless it is analogical.

iii) For **(s)pl-e̯i̯-* (beside Gmc. **splīta-*), Pokorny adduces Latv. *plītes* 'small crumbs'. But the vocalism of this form is in fact ambiguous: it can reflect either **plih$_x$-t-* or **plin-t-*, but not **ple̯i̯-t-*, unless the form is dialectal (for *-ie-* > *-ī-* in E. Latv. and SE Samogitian dialects, see e.g. Endzelin, *Comp. phonol. and morphol.* 43). Moreover, Mühlenbach–Endzelin (3.349) may be correct in connecting the form with Latv. *plîst* (: Li. *plýsti*) 'burst, crack, tear apart', which itself is apparently an analogical creation based on *plêst* (: Li. *plėšti*) 'tear, pull' (see Fraenkel s.v. *plėšti* 1). A second alleged Latv. reflex of **pl-e̯i̯-* is no more convincing, namely *pliviņa* 'detached, fluttering bark; husk'. Morphologically, the form is interpretable as a fem. diminutive in *-iņa* (for the type, see Endzelin, *Lett. Gr.* § 149 c), leading to a segmentation *pliv-iņa* (and not *pli-viņa*). The stem *pliv-*, in turn, is difficult to separate from Latv. *plivinât* 'flutter,

[22]) Pokorny sees a doubly-enlarged **(s)kr-e̯i̯-t-* in OE *mid-hridre*, OFr. *mid-hrith(ere)* 'diaphragm'; but these forms provide only the most uncertain testimony for such a stem, given the compound OE *mid(h)rif*, OFr. *midref* 'id.', clearly based on the word for 'belly', OE *hrif*, OFr. *href, hrif*. Whatever the explanation for the consonantism of OE *mid-hridre* and OFr. *mid-hrith(ere)*, the forms can hardly be dissociated from OE *mid(h)rif* and OFr. *midref*, which are entirely unrelated to IE **(s)ker(h$_x$)-* 'cut'.

float along' (see already Mühlenbach–Endzelin, 3.347), traceable to IE **pleu̯-* 'swim, float' (cf. Fraenkel s.v. *pláuti*)[23]).

For the remaining items (**drīta-*, **sprīta-*, **þwīta-*, **wlīta-*, **wrīta-*), there is no evidence at all in favor of 'intermediate' stems **dhr-ei̯-*, **spr-ei̯-*, etc.

9. The spread of the pattern within Gmc. deserves an investigation of its own; but certain preliminary observations on the problem can be made at this time.

Guxman et al. (*Sravn. gramm.* 4.307) comment on the dialectal distribution of the *lūkan* forms within Gmc., stating that such forms were 'more productive' in Ingvaeonic and Scandinavian than elsewhere. Contrast the more traditional view that the type is essentially an Ingvaeonic feature; see the comments by Matzel (referred to by Guxman et al.), Anz. f. Deut. Alt. u. Deut. Lit. 75, 1964, 56. Note further the still more restrictive view of Seebold (170, referring to MLG *drūpen* < **drūpa-*, beside OSax. *driopan*, OE *drēopan*, OHG *triofan*, OIc. *drjúpa* < **driupa-* 'tröpfen'): 'Das *ū* beim mnd. stv. ... kann sekundär sein, wie im friesisch-niederfränkischen Gebiet öfters'. But even here, Seebold is troubled by the vocalism of the weak verbs OIc. *drúpa* 'hang, let drop' and OE *drūpian* 'drip'. I cannot agree with Seebold that 'das nordische Wort steht in der Bedeutung weit ab und ist unter Umständen nicht zugehörig'; such a view seems prompted only by a wish to explain away the forms with *ū*-vocalism. As for the rare OE *drūpian*, 'ist es gänzlich unsicher, ob hier eine Länge vorliegt' (Seebold, loc. cit.). But even this observation is seriously weakened by the fact that OE *drūpian* : OIc. *drúpa* fall into a well-established pattern, according to which weak *ō*-verbs in OE (evidently secondarily) correspond to *ē*-verbs elsewhere: thus OE *ārian* : OHG *ērēn* 'ehren', OE *folgian* : OHG *folgēn* 'folgen', OE *hlinian* : OHG *hlinēn* 'lehnen', OE *hatian* : OHG *hatēn* 'hassen', and a number of others[24]). Thus for this case, the *ū*-vocalism extends

[23]) The remaining evidence for unextended **(s)pl-ei̯-* is judged 'sehr fraglich' even by Pokorny, and consists of the Ba.-Gmc. **ploi̯-no-* seen in Li. *pliẽnas* (cf. OPr. *playnis*) 'steel' and OIc. *fleinn* (m.) 'arrow, lance' (cf. OE *flān* (m. and f.) 'arrow, dart'; the OE f. *n*-stem *flā* 'id.' is apparently secondary, cf. Brunner-Sievers, *Altengl. Gr.* 225 and Campbell, *OEG* 251). The etymology depends, at least on the Gmc. side, on metaphorical interpretations such as de Vries' 'arrow' < **'abgespaltetes Stück' (s.v. *fleinn*). The Ba.-Gmc. connection itself has in fact been doubted; see Fraenkel s.v. *pliẽnas* (despite e.g. Trautmann, *Altpr. Sprachd.* 400, with earlier references).

[24]) See e.g. Brunner-Sievers, *Altengl. Gr.* 330–1 for a list of such pairs, among them OE *drūpian* : OIc. *drúpa*; for *drúpa* as a weak *ē*-verb of the type OIc. *vaka*

beyond Frisian and Low German territory, to include, precisely, all of 'Ingvaeonic', and Scandinavian as well. (See also below for more on the status of weak verbs with *ū*-vocalism.) The dialectal assessment of Guxman et al. is clearly correct in that the (strong) *ū*-verbs are relatively well-attested in Ingvaeonic and Scandinavian, often to the exclusion of East Gmc. and High German, where corresponding forms are frequently either unattested, or found with regular full-grade vocalism. It is important to note, however, that the type is attested with certainty in OHG: *brūhhan* 'gebrauchen', **tūhhan* 'tauchen'[25]), *-lūhhan* 'schließen', *sūgan* 'saugen', *sūfan* 'saufen', also *rūzan* 'schnarchen', and MHG *schlūchen* 'verschlingen' with OHG *slūh* 'Schlund'[26]). In Go., the only form of this type is *-lūkan* 'schließen', although this verb is well-attested with *ū*-vocalism throughout Gmc. (cf. OIc. *lúka* (later also *liúka*), OE *lūcan*, OFr. *lūka*, OSax. *-lūkan*, OHG *-lūhhan*). It seems clear, then, that the type is old within Gmc., which is a necessary assumption of the analysis presented above. But if e.g. **þrūt(a)-* is relatively old (cf. Ba. **trūd-*, and perhaps even Lat. *trūdō*), it must also be claimed that Go., and to a lesser extent High German, have innovated in eliminating or replacing the type; Ingvaeonic and North Gmc., on the other hand, reflect at the same time a more archaic situation, with more of the older forms preserved, and also a secondary phase of limited productivity. While this assessment is perhaps somewhat unorthodox, it is nevertheless fully consonant with the distribution of the forms within Gmc.[27]).

Another (marginally) productive pattern, however, is in fact characteristic of High German, and involves weak-verb derivatives with *ū*-vocalism apparently based on regular class II full-grades. With respect to weak derivatives in *-ū-*, Kuryłowicz observed that 'une fois que le verbe fort est du type *lūkan*, les dérivés correspondants ne font que répéter le vocalisme de la forme-base' (*Mél. Fourquet* 161–2, summarizing, in effect, the results of his earlier analysis of the weak

'wachen' (cf. OHG *wahhēn*, but again, OE *wacian*), see Noreen, *Altnord. Gr.* 349, despite Seebold's Gmc. **drūp-ō-* for both OIc. *drúpa* and OE *drūpian*, 170.

[25]) For justification of this reconstructed present form, see Seebold 157.

[26]) Note further OHG *strūten* 'betrügen' (**strūdja-*) and the weak type *strūchōn* 'durchstreifen', on which see below.

[27]) For the productivity of the pattern, note also the Schallverba **hriuta-/*hrūta-* (OIc. *hrjóta* 'roar, growl', beside OE *hrūtan* 'snore, snort', OFr. *hrūta* 'gasp', OSax. *hrūtan* 'snore', OHG *rūzan* 'snore; roar'), and **þiuta-/*þūta-* (OIc. *þjóta* 'howl; rush (of water); blare', OHG *diozan* 'roar (of water)', OE *þēotan* 'howl; rush' beside an OE variant *þūtan*).

verbs in *-ū-, Apophonie* 317–19). Several weak verbs of this type have already been seen above. Kuryłowicz (again following Wissmann, *Nomina postverb.* 126 ff.) pointed to a second type, in which weak *ū*-verbs correspond to regular class II full-grades. The following cases are attested:

OIc. *hrjóta* 'fall' : MHG *rūzen* (weak) 'move quickly'

OIc. *kliúfa,* OHG *klioban,* OE *clēofan* etc. 'split' : OHG *klūbōn* 'tear'

OIc. *striúka* 'rub' : OHG *strūchōn* 'rub out'

OIc. *hokinn* 'bent, crouched' : OIc. *húka,* MHG *hūchen* (weak) 'crouch, cower'

Go. *driusan,* OE *drēosan,* OSax. *driosan* 'fall' : OHG *trūrēn* 'be afflicted; lower the eyes', OE *drūsian* 'become heavy'[28])

It is noteworthy that such forms are best attested in High German, which appears to require altering the dialectal picture sketched above, according to which the spread of *ū*-vocalism was taken as characteristic only of Ingvaeonic and Scandinavian. But these forms can also be interpreted somewhat differently. Consider, first, the fact that most of the *lūkan* verbs have counterparts with regular full-grade vocalism (thus OE *scūfan* 'shove', beside OE *scēofan,* Go. *-skiuban,* OHG *skioban*). Second, as noted above, many of these complexes have weak-verb derivatives (thus OIc. *skúfa* (weak) : OE *scūfan,* like OIc. *drúpa* : MLG *drūpen* above, etc.). It is therefore conceivable that at least some of the weak forms like MHG *rūzen,* OHG *klūbōn, strūchōn, trūrēn* etc. above derive from strong verbs of the type **hrūta-, *klūba-* etc., although the forms with long vowel (in the present) have disappeared (or have been replaced by full-grades with short diphthong). If this is correct, then the data would provide further indirect evidence that *lūkan* forms may have been better-represented in High German than is generally granted.

As with the weak verbs, Kuryłowicz (*Mél. Fourquet* 162, cf. *Apophonie* 318) also takes nominal derivatives with long vocalism to be secondary as well. This is clearly the case with e.g. OE *rūst* 'rust' to Gmc. **riuda-* 'röten' (**h₁reu̯dh-*), cf. OSax./MHG/Sw. *rost* (as mentioned in § 5 (d) above). But this does not mean, pace Kuryłowicz,

[28]) Kuryłowicz also adduced OIc. *rjóða,* MHG *rieten* (strong) 'ravage' : OHG *rūtōn* 'id.'; but see Seebold 278 and 378 for the problem in establishing the strong verb, involving possible contamination with OIc. *hrjóða* 'empty, unload, pillage'.

that the forms pointing to Gmc. **grūt-* (see § 5 (b) above) necessarily reflect a secondary development from an aniṭ **griuta-*; here again, a Gmc. **grūta-* is a strong possibility, which may be indirectly supported (but not proved) by the BSl. comparanda, as already discussed.

10. Several final comments concern the status of the full-grades of the type **spriuta-, *þriuta-* etc. beside **sprūta-, *þrūt(a)-* etc.

These full-grades, in general, must be taken as secondary to the zero-grades, as is clear from the chronology of e.g. OIc. (also OSw.) *lúka* (: Go. *-lūkan* etc.), but also *liúka,* attested later. It is difficult to determine, however, whether such forms are simply analogical creations based on regular class II full-grades, or actual secondary full-grades in (*s*)*TReu̯H-d-*. They cannot, in any case, be taken to reflect *u*-basis full-grades (*s*)*TReH-u-d-,* despite the phonological ambiguity of a number of the forms. Counterevidence to such full-grades is provided by **sliuta-,* since a **skleh₂-u-d(e/o-)* would have yielded Gmc. ***slauta-*. Thus, if **sliuta-* is not simply analogical, it must reflect a secondary full-grade **skleu̯h₂-d-,* beside the **skluh₂-d-* of **slūta-*. Counterevidence of a similar sort is found in **þriuta-,* where a **treh₂-u-d-* should have produced ***þrauta-,* not to mention the additional complication of an unmotivated Schwebeablaut, given the *u*-basis **terh₂-u-* (not ***treh₂-u-*). Thus **þriuta-,* again, is better taken as a secondary full-grade **treu̯h₂-d-,* beside the **truh₂-d-* of **þrūt(a)-* (and perhaps Lat. *trūdō*). While both **spreh₁-u-d-* and **spreu̯h₁-d-* are phonologically possible for **spriuta-,* the absence of a **spreh₁-u-* nevertheless renders the secondary full-grade the more attractive of these two possibilities. Similar cases are found among the *i*-forms (although here, strictly speaking, it is not really possible to distinguish between full-grade and zero-grade). Thus for the **dherh₂-* which may underlie **drīta-,* there is evidence (outside Gmc.) for a stem **dhreh₂-* (particularly Gk. *θρᾰ́σσω* 'trouble, disturb'); but a **dhreh₂-i-d-* would have yielded Gmc. ***drait-,* and a **dhreh₂-i-* is otherwise unknown in any case. Note also **þwīta-*: if the comparison with Li. *tvóju* is correct, the form may represent **tu̯eu̯ih₂-d-* (or a zero-grade **tu̯ih₂-d-*), but not **tu̯eh₂-i-d-* (> Gmc. ***þwait-*), as mentioned in § 8 (f) above. An instructive (and still more ambiguous) case is Gmc. **smīta-* 'smear, smite' (OE *smītan* 'besmear', OFr. *smīta* 'smite, throw' etc., Seebold 437–8), which seems to belong with the **smeh₁-* of Gk. *σμάω* (infin. *σμῆν,* aor. *σμῆσαι*) 'smear, rub, anoint' (see *IEW* 966–7). While **smīta-* is multiply ambiguous, the traditional aniṭ interpretation **smeu̯-d-* (e.g.

Persson, *Studien* 155–6) is not a serious possibility: this would imply **sm-e̯i̯-d-* (and **sm-eh₁-* for *σμῆν*), whereas a ***sem-* (with appropriate meaning) does not exist. (Note also the comments in § 7 above concerning the alleged enlargement sequences **-e̯u̯-d-* and **-e̯i̯-d-*.) But a set interpretation of **smīta-* (for which **smih₁-d-*, **smei̯h₁-d-* and **smeh₁-i-d-* are all possible) appears to require an *i*-basis **smeh₁-i-* (beside simple **smeh₁-*), for which there is no other evidence[29]). Gmc. **smīta-*, then, is a secondary formation; if it is not purely analogical in origin, it should be taken either as a zero-grade **smih₁-d-* or a secondary full-grade **smei̯h₁-d-*, whose relationship to **smeh₁-* is similar to that of **spruh₁-d-* (and also **sprih₁-d-*, if such a form ever existed) to **sperh₁-*/**spreh₁-*[30]).

The BSl. evidence discussed above may also support the notion that at least some of the Gmc. forms are secondary full-grades, or rather that the origin and spread of the Gmc. full-grade type are to be understood in terms of actual set full-grades (*s*)*TReu̯H-d-*. Thus Li. *grū́džiu* : SCr. *grȕda*, Russ. *grúda* recover precisely **ghruhₓ-d-* : **ghreu̯hₓ-d-*, cf. Gmc. **grūt-* : **griuta-*.

According to Schmitt-Brandt (see *Vokalsystem* 68), enlarged 'long-diphthong' patterns of the type *CEH-T-*/*CEH-U-T-* (based on old *U*-bases *CEH-U-*) may have led to secondary 'infixed' forms *CEH-U-T-*, based on *CEH-* roots which were not in fact 'long-diphthong stems', i.e. which never appear as unenlarged *CEH-U-*, or in alternations *CEH-*/*CEH-U-*[31]). This hypothesis was viewed favorably by Kuryłowicz (BSL 63.2, 1968, 48–9, in a review of *Vokalsystem*), who considered it possible, again, that *CEH-U-T-* stems could have led to a derivational process involving the 'infixation' of a semivowel. But for the cases seen above, there is no evidence that a process of infixation was involved, i.e. that a (new) stem like **spreh₁-u-d-* ever existed. It is thus important to distinguish between hypothetical new full-grades *CEH-U-C-* (Schmitt-Brandt, Kuryłowicz), and other formations (zero-grades *CUH-C-*, whence secondary full-grades *CeU̯H-C-*) which do not necessarily presuppose the existence of new stems *CEH-U-C-*[32]).

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[29]) The alleged velar enlargements, mostly in the meaning 'small' or the like (Gk. *σμῑκρός* 'small' etc., *IEW* loc. cit.) are semantically somewhat distant from 'smear'; thus Pokorny's *smē[i]k-* : *smīk-* etc. cannot be used to support an *i*-basis **smeh₁-i-*. Aside from semantics, moreover, some of the forms require justification in other ways: note, for example, the circumflex intonation of Li. *smaĩgas* 'pole', which would need to be justified as a metatony, under the assumption that (preconsonantal) *CEHU-* yields acute intonation in Ba. (cf. n. 11 above). Pokorny's alleged 'long-diphthong' *u*-variant, finally, is not strongly supported by forms such as OIc. *móask* 'become digested' or Norw. *mugg* 'sawdust': the forms are not exactly comparable semantically; they lack the initial *s-* (*s*-mobile is not otherwise attested for this root, apart from Gk. (*σ*)*μικρός*, which is equivocal in any case, owing to the problematic double treatment of IE **sm-* in Gk.); and they are phonologically ambiguous. (For *mugg*, cf. the comments on Norw. *grugg*, § 6 (b) above.)

[30]) The Gmc. short-vowel zero-grades, as in the nouns OHG (m.) *smiz* 'spot, smear' and OE (f.) *smitte* 'stain', need not be old: *smiz* (< **smita-*) can be a concretized deverbative action-noun, and *smitte* (< **smit-jōn*), in turn, can reflect a denominative formation to **smita-*. Arm. *mic* 'mud, mire' is taken as **smi-d-i̯o-* by Pokorny, but could also derive from a **smeh₁-d-i̯o-*. The latter could then be compared with Hom. *σμῶδιξ* 'welt, bruise', which seems to presuppose an *o*-grade **smoh₁-d-(o-)*. The connection with CS *směděъ 'μαυρός, μέλας'* (SCr. *smẽd* 'brunette'), and perhaps Russ. river names based on a *ū*-stem **smẽdy*, is regarded as uncertain by Vasmer (s.v. *Smedva*), and would again involve the ambiguity of diphthongal versus non-diphthongal interpretations for the stem.

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[31]) Similarly – and conversely, in a sense – old *CEH-U-T-* forms could have led to 'artificial' *U*-less forms *CEH-T-*.

[32]) For further discussion of secondary full-grades (and related problems), see my treatment in *IE *-d-* 257–70.

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